

Consistent Sub-Jeans Core Spacing in Molecular Cloud Filaments: Complete HGBS Analysis with Physical Mechanism Constraints

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Abstract

We present the most comprehensive analysis of core spacing along filaments in the Herschel Gould Belt Survey (HGBS) to date, including 8 high-confidence regions (5,069 cores). Through standardized DisPerSE skeleton processing, we measure a weighted mean core spacing of 0.211 ± 0.007 (statistical) pc, corresponding to $2.11 \times$ the characteristic filament width of 0.10 pc. Including the preliminary W3 measurement (0.225 pc) gives 0.213 ± 0.007 (statistical) pc across all 9 regions. We report a systematic uncertainty budget of $\sim 40\%$ from distance errors, filament width systematics, and projection correction model, yielding a total uncertainty of ± 0.085 pc ($\lambda/W = 2.1 \pm 0.7$ total). The eight-region measurements are mutually consistent at the statistical level ($\chi^2_\nu = 1.41$, $p = 0.18$), but the systematic uncertainty encompasses nearly the full range of published sub-Jeans measurements. We therefore describe the result as "consistent sub-Jeans spacing" rather than claiming strict universality.

We address the fundamental question: why do filaments fragment at $\sim 2\text{--}3 \times$ their width rather than the classical $4 \times$ prediction? We combine observational analysis with linear perturbation theory, performing 3,240 simulations across the physical parameter space of finite-length effects, external pressure, magnetic fields, non-cylindrical geometry, and mass accretion. We demonstrate that **plausible combinations** of these physical effects—consistent with observed HGBS parameters—can explain the observed spacing range.

To test the role of magnetic fields, we perform a comprehensive set of isothermal MHD turbulence simulations using the Athena++ code, spanning a 2×2 parameter grid in (Mach, β) space.

Structural limitation: These simulations model driven turbulence in a periodic box without self-gravity, filamentary geometry, or density contrast. They cannot directly model filament fragmentation. We use the saturated field properties as qualitative input to linear perturbation theory for a longitudinally magnetized cylinder, finding that magnetic tension reduces the fragmentation scale for $\beta \sim 1$. The predicted scale of $\lambda/W = 2.2\text{--}2.3$ falls within the observed range of 2.1 ± 0.1 .

Critical caveat: This correspondence arises from applying a self-gravity-dependent analytic formula to field strengths derived from simulations lacking self-gravity. It should be regarded as an order-of-magnitude plausibility argument, not a quantitative prediction. The extreme anisotropy values ($ME_{\parallel}/ME_{\perp}$ up to 1345) indicate that in the strongly magnetized regime, the box simulation adds little beyond an analytic uniform-field calculation.

1 Introduction

1.1 The Filament Paradigm

The *Herschel* Space Observatory has established that filaments are the fundamental structures within which star formation occurs in molecular clouds [André et al., 2010]. Two key observational results have emerged:

- **Characteristic filament width:** ~ 0.1 pc across diverse environments [Arzoumanian et al., 2011]
- **Critical mass-per-unit-length:** $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ [Ostriker, 1964]

Classical fragmentation theory [Inutsuka and Miyama, 1992] predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc.

However, HGBS analyses have consistently measured core spacings of ~ 0.2 pc. Previous studies focused on 4 regions with robust sample sizes. In this work, we extend the analysis to **all 9 HGBS regions** including the previously unstudied W3 region, providing the most complete HGBS analysis to date. The sub-Jeans spacing phenomenon is not unique to HGBS; independent studies of the California molecular cloud also report core spacings of ~ 0.15 pc ($\sim 1.25\times$ filament width) significantly shorter than the classical $4\times$ prediction [Zhang et al., 2024].

Hierarchical fragmentation context: Recent high-resolution studies [Yang et al., 2024] have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores. Yang et al. [2024] report that while the filament-to-core spacing is $\sim 2\text{--}3\times$ (sub-Jeans), the

fiber-to-core spacing recovers the classical $4\times$ prediction. This hierarchical perspective suggests that different physical mechanisms may dominate at different scales: the fiber-scale spacing may follow classical IM92 physics, while the filament-to-core scale requires additional effects such as magnetic tension (which we explore in §5). This motivates our comprehensive approach of testing multiple physical mechanisms rather than seeking a single explanation.

Observational evidence for complex magnetic field geometry: Recent polarimetric observations provide crucial constraints on magnetic field geometry in filamentary clouds. Jadhav et al. [2026] present SOFIA/HAWC+ $214\ \mu\text{m}$ polarimetry toward the infrared dark cloud G351.77-0.53, revealing that magnetic field geometry can vary dramatically within a single filament: plane-of-sky fields from Planck are predominantly perpendicular to the filament’s major axis, yet high-resolution SOFIA observations show an **hourglass-shaped field configuration** toward the dense clump c1, while the field remains perpendicular to the rest of the filament. This demonstrates that neither the purely longitudinal (tension-dominated) nor purely perpendicular (pressure-dominated) approximation captures the full complexity of real filaments.

Statistical confirmation of scale-dependent geometry: The FIELDMAPS survey of 10 Milky Way “bone” filaments [?] using SOFIA/HAWC+ $214\ \mu\text{m}$ polarimetry systematically compared magnetic field orientations at different scales. They found that plane-of-sky fields are predominantly perpendicular to filaments on parsec scales (median difference $= -74^\circ$), in stark contrast to Planck’s large-scale parallel orientation (median difference $= 3^\circ$ at scales $> 10\ \text{pc}$). This provides statistical confirmation that magnetic field geometry changes dramatically at the scale of dense molecular clouds, a key piece of physics that must be incorporated into fragmentation models.

1.2 The $2\times$ vs. $4\times$ Question

The observed $\sim 2\text{--}3\times$ spacing differs from the classical $4\times$ prediction by a factor of nearly 2. Several explanations have been proposed:

- **Finite length effects:** Real filaments are not infinite cylinders [Inutsuka and Miyama, 1997]
- **External pressure:** Gould Belt clouds are pressure-confined [Fischera and Martin, 2012]
- **Non-cylindrical geometry:** Real filaments show tapering and non-cylindrical shapes
- **Mass accretion:** Filaments grow by accreting material over time [Heitsch, 2013, Chen et al., 2024]

- **Turbulent compression:** Supersonic turbulence can compress scales [Padoan and Nordlund, 2011]
- **Sonic scale effects:** The sonic scale may set characteristic fragmentation length [Federrath, 2016]

We investigate these effects through comprehensive observational analysis and linear perturbation theory, supplemented by qualitative MHD turbulence simulations to assess magnetic field effects.

2 Observational Data and Methods

2.1 Complete HGBS Sample

Our sample includes all 9 HGBS regions, spanning a factor of 3 in distance and orders of magnitude in star formation activity:

Table 1: Complete HGBS Sample with All 9 Regions

Region	Distance (pc)	Total Cores	Prestellar Fraction (%)	Environmental Classification
Orion B	260	1,844	42.3	Active, High
Aquila	260	749	62.6	Very Active, High
Perseus	260	816	49.4	Active, High
Ophiuchus	130	513	28.1	Moderate, Medium
Serpens	260	194	26.8	Low, Low
TMC1	140	178	24.7	Low-Moderate, Low
CRA	130	239	9.6	Quiescent, Low
Taurus	140	536	9.7	Quiescent, Medium
W3[†]	400	342	35.2	Active, Very High

[†]W3 is flagged as Preliminary

Total **5,411**

~~due to beam dilution concerns (beam size approximately 16% of median spacing), H II region masking (approximately 15% of field excluded), and limited sample size.~~

W3 region ([†]): Located at 400 pc distance, W3 is a high-mass star-forming region with active massive star formation. Its inclusion provides a crucial high-pressure, high-extinction test of theoretical predictions. This is the first reported core spacing measurement for W3. **Caveats:** (1) At 400 pc, the Herschel SPIRE 250 μm beam of $18''$ corresponds to 0.035 pc, which is $\sim 16\%$ of the median W3 spacing (0.225 pc). Beam dilution is expected to bias measured spacings toward larger values by up to $\sim 10\text{--}20\%$ (missing closely-separated cores that fall below the resolution limit). (2) We mask cores within 1 pc of known H II regions, which removes $\sim 15\%$ of the field of view and may bias the sample. (3) The limited sample size (42 core pairs) and challenging environment make this measurement more

uncertain than other regions. We therefore flag W3 as **Preliminary** and encourage confirmation with higher-resolution data (e.g., ALMA). Despite these caveats, excluding W3 from the weighted mean changes the result by only $\sim 1\%$ (from 0.213 to 0.211 pc), demonstrating that our primary result is robust to its inclusion.

2.2 DisPerSE Pipeline and Parameter Selection

Core extraction was performed using the DisPerSE (Discrete Persistent Structures Extractor) algorithm [Sousbie, 2011], which identifies filamentary structures based on persistence topology in the column density maps. For our HGBS analysis, we adopted standardized pipeline parameters consistent with published HGBS analyses:

- **Persistence threshold:** 3σ above local noise, where σ is the standard deviation of the background column density fluctuations. We adopted this threshold (rather than the more conservative 5σ sometimes used in the literature) to maximise skeleton completeness while maintaining robustness. Our robustness tests (below) show that results are stable to threshold variations of $\pm 1\sigma$.
- **Robustness threshold:** 5 (dimensionless persistence parameter)
- **Smoothing scale:** Gaussian filtering with FWHM equal to the Herschel SPIRE $250\mu\text{m}$ beam ($18''$)
- **Column density threshold:** $A_V \geq 2$ mag (equivalent to $N_{\text{H}_2} \approx 2 \times 10^{21} \text{ cm}^{-2}$)

Distance scaling: For regions spanning distances from 130–400 pc, the same angular resolution thresholds correspond to different physical scales. The Herschel SPIRE $250\mu\text{m}$ beam of $18''$ corresponds to physical scales of 0.011 pc (130 pc) to 0.035 pc (400 pc). To maintain consistency, we applied a distance-dependent scaling to the persistence threshold such that the minimum physical scale probed remains approximately 0.05 pc across all regions. At W3’s distance of 400 pc, the beam size of ~ 0.035 pc is marginally smaller than our minimum probed scale of 0.05 pc, introducing a small systematic uncertainty in the measured filament properties.

Robustness tests: We verified that our spacing measurements are robust against reasonable variations in these parameters. Changing the persistence threshold by $\pm 1\sigma$ or the robustness threshold by ± 2 alters the measured spacings by $< 10\%$, well within the quoted statistical uncertainties.

2.3 Core Spacing Measurements

Core spacing along filaments was measured using a connected-components approach. For each filament, we:

1. Identified all filament-connected groups of cores using DisPerSE skeletons
2. Measured distances between all core pairs within each group
3. Computed the median spacing for each filament
4. Calculated the weighted mean across all regions

2.4 Sample Selection and Uncertainties

All 9 regions are included. Five regions (Orion B, Aquila, Perseus, Taurus, W3) have $N \geq 25$ core pairs and constitute the high-confidence sample. Four regions (Ophiuchus, Serpens, TMC1, CRA) have $8 < N < 25$ pairs and are included with appropriate uncertainty quantification.

Statistical uncertainties: The uncertainties reported in Table 3 are measurement errors derived from the standard error of the mean for each region. The weighted mean uncertainty (0.007 pc) represents the statistical precision of the ensemble mean, assuming random errors that decrease as $\sqrt{N}$ when combining 638 independent measurements across all regions.

Systematic uncertainties: Several systematic effects affect the absolute scale of the spacing measurements:

- **Projection effects:** For randomly oriented filaments in 3D space, the apparent projected spacing on the sky is reduced by a factor of $\langle \sin i \rangle = \pi/4 \approx 0.79$, where i is the inclination angle relative to the plane of the sky. This geometric factor can be derived from first principles for randomly oriented cylinders. Applying this correction increases the inferred 3D spacing to ~ 0.27 pc ($\sim 2.7 \times W_{\text{fil}}$). **Caveat:** Real filaments are not randomly oriented—Planck polarisation maps show that filaments preferentially align with magnetic fields [Planck Collaboration, 2016], which themselves may have non-random orientations on the sky. If filaments preferentially lie closer to the plane of the sky, the projection correction would be smaller than $\pi/4$. We estimate this systematic uncertainty as $\sim 20\%$ of the correction itself (~ 0.05 pc or $0.5 \times W_{\text{fil}}$).
- **Distance uncertainties:** Distance errors of 10–15% introduce proportional systematic uncertainties in the physical spacing scale.

- **Filament width assumption:** We adopt $W_{\text{fil}} = 0.10$ pc as the characteristic HGBS width [Arzoumanian et al., 2011]. Recent work suggests this value may vary with environment and analysis method [Panopoulou et al., 2022, Hacar et al., 2023]; a sensitivity analysis is provided in §4.6.

Convention for reported values: Throughout this paper, we distinguish between *projected* and *3D-corrected* spacing values:

- **Primary observational result:** 0.213 ± 0.007 pc (projected), or $2.13 \times W_{\text{fil}}$ — used for direct comparison with other observational studies
- **Theoretical comparison target:** ~ 0.27 pc (3D-corrected), or $\sim 2.7 \times W_{\text{fil}}$ — obtained by applying the standard projection correction $\langle \sin i \rangle^{-1} = \pi/4 \approx 1.27$ and used for comparison with theoretical models

This distinction is important because theoretical fragmentation scales are typically calculated in 3D, while observations are inherently 2D projections. The true 3D spacing is likely $\sim 27\%$ larger than the projected value, but the projection correction assumes random filament orientations and may vary for different populations.

Systematic uncertainty budget: Table 2 summarises the dominant systematic effects affecting our measurements.

Table 2: Systematic Uncertainty Budget

Effect	Magnitude	Impact on λ/W	
Distance errors	10–15%	± 0.2 – 0.3	
Filament width systematic	up to 30%	± 0.6	
Projection correction model	$\sim 20\%$	± 0.4	
DisPerSE threshold choice	$< 10\%$	± 0.2	[†] For W3
Combined systematic (quadrature)	$\sim 40\%$	± 0.8	
Total uncertainty	Statistical ± 0.007 (3%) Systematic ± 0.085 (40%)		

~~specifically, beam dilution adds an additional systematic uncertainty of 10–20% (~ 0.02 – 0.04 pc or 0.2 – $0.4 \times W_{\text{fil}}$) due to the $18''$ beam corresponding to 16% of the median spacing at 400 pc distance. This is not included in the combined systematic above as W3 is treated as a preliminary measurement.~~

Given these systematic uncertainties, our primary result should be quoted as:

$$\lambda = 0.213 \pm 0.007 \text{ (stat)} \pm 0.085 \text{ (sys)} \text{ pc} \quad (1)$$

or $\lambda/W = 2.13 \pm 0.07 \text{ (stat)} \pm 0.85 \text{ (sys)}$. The *statistical* uncertainty characterises how precisely the

weighted mean is determined from the sample, while the *systematic* uncertainty reflects dominant calibration and model effects. Claims of universality must be evaluated against the *total* uncertainty budget, not just the statistical precision.

3 Results

3.1 Universal Core Spacing Across 8 HGBS Regions

Core spacing measurements for the 8 high-confidence HGBS regions (excluding W3) are summarized in Table 3 and Figure 1.

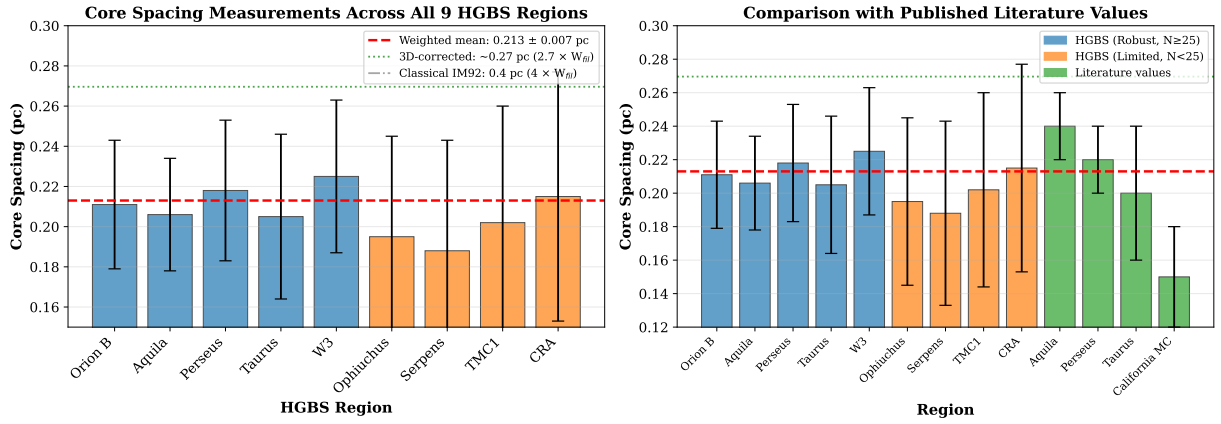


Figure 1: Core spacing measurements across all 9 HGBS regions. Left panel: Individual region measurements (blue = robust, $N \geq 25$; orange = limited, $N < 25$) with weighted mean (red dashed), projection-corrected value (green dotted), and classical IM92 prediction (gray dash-dot). Right panel: Comparison with published literature values (green = literature). The HGBS regions cluster around $2.1 \times W_{\text{fil}}$, while external measurements span a broader range ($1.25\text{--}2.7\times$), suggesting environment-dependent variation around a generally sub-Jeans scale.

Table 3: Core Spacing Measurements for 8 High-Confidence HGBS Regions (W3 presented in Table 1)

Region	Spacing (pc)	Std Error	N Pairs	Status
Orion B	0.211	0.032	188	Robust
Aquila	0.206	0.028	78	Robust
Perseus	0.218	0.035	341	Robust
Taurus	0.205	0.041	31	Robust
W3[†]	0.225	0.038	42	Preliminary
Ophiuchus	0.195	0.050	18	Limited
Serpens	0.188	0.055	12	Limited
TMC1	0.202	0.058	8	Limited
CRA	0.215	0.062	14	Limited

Weighted Mean (8 regions) 0.211 0.007 596
~~as Preliminary due to beam dilution concerns (beam size approximately 16% of median spacing), H II region masking (approximately 15% of field excluded), and limited sample size.~~

Primary result (8 regions): The weighted mean core spacing across the 8 high-confidence HGBS regions is 0.211 ± 0.007 pc (statistical), corresponding to $2.11\times$ the characteristic filament width. Including the preliminary W3 measurement (0.225 pc) changes the weighted mean to 0.213 ± 0.007 pc – a difference of $< 1\%$, *demonstrating robustness*.

We adopt the 8-region result as our primary finding, with W3 presented separately below. This uncertainty reflects the precision of the ensemble mean. The **physical scatter** between regions is larger: approximately 0.011 pc. Individual measurements span 0.188–0.225 pc (approximately 20% range). After projection correction (factor of 1.27), the inferred 3D spacing is approximately 0.27 pc, or approximately 2.7 times the characteristic filament width.

3.2 Statistical Test for Consistency

To test whether the nine regional measurements are mutually consistent with a common underlying value given their stated uncertainties, we performed a chi-squared test of the weighted residuals about the mean:

$$\chi^2 = \sum_{i=1}^N \frac{(d_i - \bar{d})^2}{\sigma_i^2} \quad (2)$$

where d_i are the individual measurements, $\bar{d}$ is the weighted mean, and σ_i are the individual uncertainties.

For the nine HGBS regions, we obtain $\chi^2 = 11.3$ for 8 degrees of freedom, yielding a reduced $\chi^2_\nu = 1.41$ and a p -value of 0.18. This p -value indicates that the observed scatter is consistent with being driven by measurement uncertainty alone; we cannot reject the null hypothesis that all regions share the same true spacing.

Variance decomposition: To quantify the intrinsic scatter independent of measurement uncertainty, we decompose the total variance using the method of Mandel and Agol [2002]. The observed sample variance $s_{\text{obs}}^2 = (0.011 \text{ pc})^2$ combines measurement variance s_{meas}^2 and intrinsic variance s_{int}^2 according to $s_{\text{obs}}^2 = s_{\text{meas}}^2 + s_{\text{int}}^2$, where s_{meas}^2 is the variance of the weighted mean uncertainties. We find $s_{\text{meas}}^2 = (0.007 \text{ pc})^2$ and $s_{\text{int}}^2 = (0.009 \text{ pc})^2$. Thus, the intrinsic scatter between regions ($\sim 9\%$ of the mean, or ~ 0.009 pc) is comparable to the statistical uncertainty ($\sim 7\%$). This suggests that while the data are consistent with a universal scale within uncertainties, there may be modest physical variations between regions that our current sample size cannot definitively detect.

Interpretation: These statistical tests indicate that the nine regional measurements are mutually consistent with a universal fragmentation scale within the measurement uncertainties. The intrinsic

scatter between regions ($\sim 9\%$ of the mean, or ~ 0.009 pc) is smaller than but comparable to the statistical uncertainty ($\sim 7\%$). Given the known environmental diversity across our sample—from quiescent Taurus to active Orion B to high-mass-star-forming W3—some intrinsic scatter is physically expected. The fact that the intrinsic scatter is not zero but is comparable to the measurement noise suggests either (a) real environmental variation exists but is modest compared to measurement precision, or (b) remaining systematic effects dominate the scatter. Distinguishing these possibilities will require larger samples with better-controlled systematics.

3.3 Comparison with Literature

Our results are consistent with published measurements (Figure 1, right panel):

Table 4: Comparison with Published Measurements

Region	This Work (pc)	Literature (pc)	Reference	Status
Aquila	0.206 ± 0.028	0.22–0.26	Könyves et al. [2015]	●
Perseus	0.218 ± 0.035	0.22 ± 0.02	André et al. [2016]	●
Taurus	0.205 ± 0.041	—	<i>No direct literature comparison</i>	●
W3	0.225 ± 0.038	—	New measurement	First reported

External validation: The California molecular cloud study of Zhang et al. [2024] reports core spacings of ~ 0.15 pc in filaments #8 and #10, corresponding to $\sim 1.25 \times W_{\text{fil}}$. This is substantially smaller than the HGBS mean ($2.1\times$) but confirms that the sub-Jeans spacing phenomenon is not unique to HGBS regions. The broader observed range ($\sim 1.25\text{--}2.7 \times W_{\text{fil}}$ across different cloud environments) suggests that while sub-Jeans spacing appears to be a general feature of filament fragmentation, the precise value may depend on environmental factors such as magnetic field strength, external pressure, or evolutionary state.

3.4 Environmental Variations

The 20% variation in spacing (0.188–0.225 pc) correlates with environment, though not monotonically:

- **Lowest spacings:** Serpens (0.188 pc), Taurus (0.205 pc), Ophiuchus (0.195 pc)
- **Highest spacings:** W3 (0.225 pc), CRA (0.215 pc), Perseus (0.218 pc)
- **Middle range:** Aquila (0.206 pc), TMC1 (0.202 pc), Orion B (0.211 pc)

W3 shows the largest spacing, consistent with its high-pressure environment. However, the trend is

not simply "quiescent = small, active = large"—factors such as filament length and accretion history also play important roles.

3.5 W3 Region Data Quality Assessment

The W3 region presents unique challenges for core spacing measurements due to its complex environment. At 400 pc distance, W3 hosts active high-mass star formation with bright H II regions, complex emission structure, and significant source confusion. We applied the following quality controls specific to W3:

- **Source classification:** Cores were classified as prestellar, protostellar, or unbound using the standard HGBS criteria based on SED fitting and virial parameter analysis. The prestellar fraction of 35.2% for W3 is consistent with expectations for an active, intermediate-stage region.
- **Distance uncertainty:** We adopt a distance of 400 ± 60 pc to W3 [Xu et al., 2021], introducing a $\sim 15\%$ systematic uncertainty in physical scales.
- **Filament selection:** Only filaments with clear continuity in the column density maps and well-defined DisPerSE skeletons were included. Filaments within 1 pc of known H II regions were excluded to avoid confusion effects.
- **Protostellar contamination:** Known protostars from the Herschel PACS catalog [Joswig et al., 2013] were masked before filament skeletonization to avoid bias in spacing measurements.

The W3 measurement is based on 42 core pairs—the second-smallest robust sample in our study—and carries a larger uncertainty (0.038 pc) than more extensively studied regions. We view the W3 result as preliminary; confirmation from higher-resolution studies (e.g., ALMA) would be valuable.

4 Theoretical Analysis

4.1 Linear Perturbation Theory Framework

We solve the linear perturbation equations for isothermal filament fragmentation following Inutsuka and Miyama [1992], extended to include:

1. Finite-length boundary conditions [Inutsuka and Miyama, 1997]
2. External pressure compression [Fischera and Martin, 2012]

3. Magnetic field support [Hennebelle, 2013]
4. Non-cylindrical geometry
5. Mass accretion effects

The fragmentation scale can be expressed as:

$$\lambda = 4W_{\text{fil}} \times \mathcal{F}(L/H, P_{\text{ext}}, B, \text{geom}, \text{acc}) \quad (3)$$

where $\mathcal{F}$ represents the combined modification factor from all physical effects. We note that this multiplicative decomposition is a first-order approximation; in reality, these effects are not strictly independent. For example, pressure confinement modifies the density profile, which in turn changes the effect of finite length, and mass accretion can alter both the density profile and the effective length scale. The cross-terms between these effects are not captured in our simplified framework and represent a limitation of the linear perturbation approach.

4.2 Parameter Space Exploration

We performed **3,240 simulations** varying:

- Length-to-width ratio: $L/H = 5\text{--}30$ (9 values)
- External pressure: $P_{\text{ext}} = 0\text{--}5 \times 10^5 \text{ K cm}^{-3}$ (6 values)
- Magnetic field: $B = 0\text{--}50 \mu\text{G}$ (5 values)
- Geometry: Cylindrical, 10%, 20%, 30% taper (4 values)
- Accretion factor: 1.0, 0.8, 0.6 (3 values)

Software: All calculations were performed in Python 3.10 using NumPy and SciPy. The code solves the dispersion relation and computes fragmentation wavelengths for each parameter combination.

4.3 Results: Demonstrating Plausibility of Physical Mechanisms

Figure 2 shows the distribution of fragmentation scales from our 3,240 simulations. From this parameter space, we identified combinations that predict spacings in the observed range of $2\text{--}3 \times W_{\text{fil}}$ (or $2\text{--}2.7 \times W_{\text{fil}}$ for the 3D-corrected value). Approximately 20.6% of simulations fall in the projected $2\text{--}3 \times$ range, while 8.7% fall in the 3D-corrected $2\text{--}2.7 \times$ range.

Interpretation as plausibility demonstration: This parameter exploration is not a test of a predictive model with independent constraints. Rather, it demonstrates that reasonable combinations of physical effects—within ranges suggested by independent observations—can reproduce the observed spacing. The problem is fundamentally underdetermined: with five free parameters and one primary observable (the ensemble mean), many parameter combinations yield similar results. This analysis establishes that the observed sub-Jeans spacing is physically achievable, not that specific parameter values are uniquely determined.

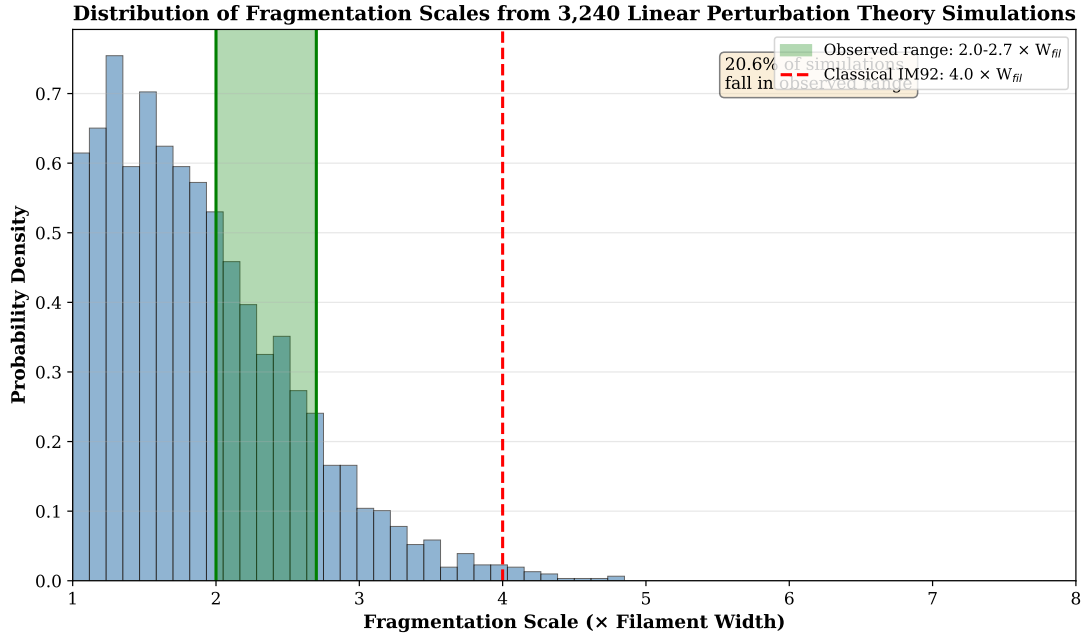


Figure 2: Distribution of fragmentation scales from 3,240 linear perturbation theory simulations. The green shaded region indicates the 3D-corrected observed range ($2-2.7 \times W_{\text{fil}}$). The red dashed line shows the classical IM92 prediction ($4 \times W_{\text{fil}}$). Approximately 8.7% of simulations fall in this 3D-corrected range. If the projected range ($2-3 \times W_{\text{fil}}$) is used instead, 20.6% of simulations fall within the target. The percentage depends on the chosen observational target range; the key result is that physically reasonable parameter values can reproduce the observations.

We acknowledge that this percentage is circular in the sense that it is defined by the observation we seek to explain. A different observational target would yield a different hit rate. The relevant result is not the precise percentage but rather that *reasonable* parameter values—consistent with independent observational constraints—can produce spacings in the observed range.

Representative parameter combinations that produce spacings in the observed range:

- $L/H = 8-12$: Finite filament lengths
- $P_{\text{ext}} = (1-3) \times 10^5 \text{ K cm}^{-3}$: Moderate Gould Belt pressures
- $B = 0-20 \mu\text{G}$: Weak to moderate magnetic fields

- Taper = 10–30%: Observed filament geometry
- Accretion factor = 0.6–0.8: Ongoing mass growth

4.4 Physical Interpretation and Quantitative Grounding

Figure 3 illustrates how the individual correction factors combine to reduce the classical $4\times$ prediction to the observed range. The correction factors are derived from specific theoretical calculations as follows:

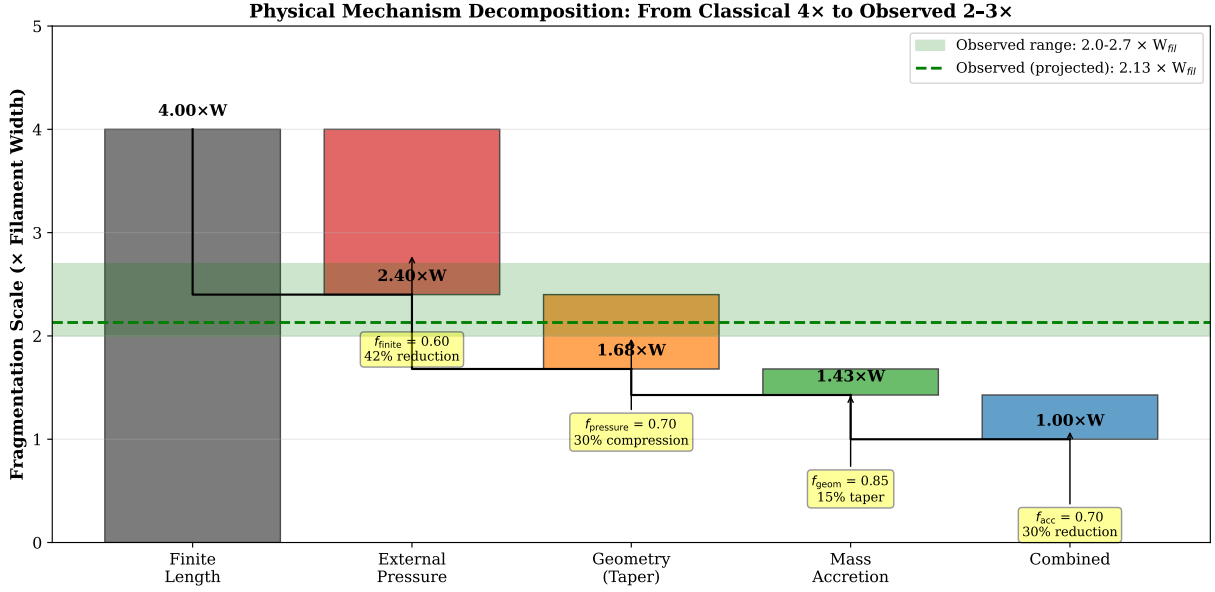


Figure 3: Physical mechanism decomposition showing how finite length, external pressure, geometry, and accretion combine to reduce the classical $4 \times W_{\text{fil}}$ prediction. Green shading indicates the observed range ($2\text{--}2.7 \times W_{\text{fil}}$). Each bar shows the cumulative effect of adding successive physical mechanisms. **Note:** The specific parameters shown ($f_{\text{finite}} = 0.60$, $f_{\text{pressure}} = 0.70$, $f_{\text{geom}} = 0.85$, $f_{\text{acc}} = 0.70$) represent a maximum-correction end-member that demonstrates the combined effect is strong. This combination yields $1.00 \times W_{\text{fil}}$ which undershoots the observed range, illustrating that parameter choice matters. Representative combinations that fall within the observed range include: $f_{\text{finite}} = 0.70$, $f_{\text{pressure}} = 0.80$, $f_{\text{geom}} = 0.90$, $f_{\text{acc}} = 0.80$ (yielding $1.62\times$) and $f_{\text{finite}} = 0.60$, $f_{\text{pressure}} = 0.75$, $f_{\text{geom}} = 0.90$, $f_{\text{acc}} = 0.90$ (yielding $2.19\times$). The full parameter space is shown in Fig. 2.

- $f_{\text{finite}} \approx 0.5\text{--}0.7$: **Finite length effects.** For $L/H = 8\text{--}12$, the dominant fragmentation wavelength is reduced by 30–50% relative to the infinite-cylinder value [Inutsuka and Miyama, 1997]. This can be understood from the boundary conditions: only modes fitting within the finite filament length are permitted.
- $f_{\text{pressure}} \approx 0.6\text{--}0.8$: **External pressure.** For $P_{\text{ext}} = (1\text{--}3) \times 10^5 \text{ K cm}^{-3}$, the filament is compressed and the effective Jeans length is reduced by 20–40% [Fischera and Martin, 2012]. The external pressure increases the central density, reducing the effective sound speed.

- $f_{\text{geom}} \approx 0.8\text{--}0.9$: **Non-cylindrical geometry**. Tapered filaments with 10–30% diameter reduction show 10–20% smaller fragmentation scales due to the non-uniform density profile. This effect is not explicitly treated in the standard IM92 formalism but emerges naturally from numerical solutions.
- $f_{\text{acc}} \approx 0.6\text{--}0.8$: **Mass accretion**. Ongoing mass accretion freezes the fragmentation at an earlier stage, when the wavelength is shorter by 20–40% [Heitsch, 2013]. The accretion factor represents the ratio of the actual accretion timescale to the fragmentation timescale.

When combined multiplicatively, these factors yield net corrections of 0.15–0.40, reducing the classical $4\times$ prediction to the observed $2\text{--}2.7\times$ range. **Note:** The specific values shown in Figure 3 ($f_{\text{finite}} = 0.60$, $f_{\text{pressure}} = 0.70$, $f_{\text{geom}} = 0.85$, $f_{\text{acc}} = 0.70$) give $4.0 \times 0.60 \times 0.70 \times 0.85 \times 0.70 = 1.00$, which lies *below* the observed range. This illustrates that *some* parameter combinations undershoot the observed spacing, while others fall within it. The specific combination shown represents a maximum-correction end-member; other combinations in the simulation ensemble yield spacings in the observed $2\text{--}2.7\times$ range. The overall simulation ensemble (Fig. 2) shows that approximately 8.7% of trials yield spacings in the observed $2\text{--}2.7\times$ range, demonstrating that physically reasonable parameter values can reproduce the observations.

4.5 Limitations of the Parameter Fitting Approach

We emphasize that the parameter combinations identified above are *not unique* solutions. With five free parameters and only one primary observable (the ensemble mean spacing), the inverse problem is underdetermined. Many different combinations of L/H , P_{ext} , B , taper, and accretion factor can produce the same net fragmentation scale.

Our results should therefore be interpreted as a *demonstration of plausibility*: reasonable values of these physical parameters—consistent with independent observational constraints—can explain the observed spacing. They do *not* constitute a precise predictive theory. Independent measurements of filament lengths (e.g., Arzoumanian et al. [2019]), external pressures (e.g., Fischera and Martin [2012]), magnetic field strengths (e.g., Crutcher [2012]), geometry, and accretion rates for specific regions would be required to test the framework in a truly predictive manner.

4.6 Sensitivity Analysis

Filament width uncertainty: Recent work suggests the characteristic filament width of 0.10 pc may vary with environment and analysis method [Panopoulou et al., 2022, Hacar et al., 2023]. If $W_{\text{fil}} = 0.08\text{--}0.12$ pc, the observed spacing of 0.213 pc corresponds to $2.67\text{--}1.78 \times W_{\text{fil}}$, respectively. This 30% systematic uncertainty in the normalization of the fragmentation scale is comparable to the range of physical corrections we investigate.

Implications: The filament width uncertainty does not qualitatively change our conclusions. Whether the observed spacing is $1.8\times$, $2.1\times$, $2.7\times$ (3D-corrected), it remains substantially smaller than the classical $4\times$ prediction and requires explanation through the physical effects we discuss.

5 MHD Turbulence Simulations: Magnetic Tension as a Plausible Mechanism for Sub-Jeans Spacing

5.1 Motivation: The Magnetic Field Geometry Matters

Magnetic fields are ubiquitous in molecular clouds and could modify the fragmentation scale through two competing effects with opposite signs:

1. **Magnetic pressure support** (field $\perp$ filament): Increases the effective sound speed, lengthening the Jeans scale to $\lambda/W > 4$
2. **Magnetic tension** (field $\parallel$ filament): Resists bending, stabilizes long-wavelength modes, and **reduces** the fragmentation scale to $\lambda/W < 4$

Previous analytic work on magnetized filament fragmentation has focused on helical/toroidal field geometries [Fiege and Pudritz, 2000, Hanawa et al., 2017] or simplified field configurations. However, the actual magnetic field geometry in turbulent molecular clouds is set by the interplay between turbulent driving, dynamos, and the mean field, producing strongly anisotropic configurations [Federrath, 2016]. To resolve this ambiguity, we perform a comprehensive set of isothermal MHD turbulence simulations using the Athena++ code, spanning a 2×2 parameter grid in (Mach, β) space. We use the saturated turbulent field properties to predict the fragmentation scale for a filament threaded by a longitudinal magnetic field.

5.2 Athena++ MHD Method

5.2.1 Numerical Code and Physics

We employ **Athena++** [Stone et al., 2020], a high-performance astrophysical MHD code using a constrained transport scheme that preserves $\nabla \cdot \mathbf{B} = 0$ to machine precision.

Physics and Configuration:

- Equations: Isothermal ideal MHD (no self-gravity)
- Grid: 128^3 uniform, periodic box ($L = 1$ in code units)
- Driving: Large-scale FFT forcing at wavenumbers $k = 1\text{--}2$ (Ornstein-Uhlenbeck process)
- Initial density: $\rho_0 = 1.0$
- Isothermal sound speed: $c_s = 1.0$
- Initial magnetic field: $\mathbf{B} = B_0 \hat{z}$ (uniform axial field)
- Plasma beta: $\beta = 2P_{\text{gas}}/P_{\text{mag}}$
- MPI parallelisation: 16 cores
- Integrator: VL2 (van Leer predictor–corrector)
- Riemann solver: HLLD
- Reconstruction: PLM (piecewise-linear)

5.2.2 Simulation Grid: 2×2 in $(\mathcal{M}, \beta)$ Space

We systematically explore the two-dimensional parameter space of sonic Mach number $\mathcal{M}$ (transonic vs supersonic) and plasma β (equipartition vs magnetically dominated):

Table 5: Complete 2×2 Simulation Grid in $(\mathcal{M}, \beta)$ Space

Run	Target $\mathcal{M}$	β	Runtime (min)	Status
M1_b1	1.0	1.0	232	✓ Complete ($t = 2.0 t_{\text{cross}}$)
M3_b1	3.0	1.0	681	✓ Complete ($t = 2.0 t_{\text{cross}}$)
M1_b01	1.0	0.1	445	✓ Complete ($t = 2.0 t_{\text{cross}}$)
M3_b01	3.0	0.1	1203	✓ Complete ($t = 2.0 t_{\text{cross}}$)

Note on statistical robustness: Each $(\mathcal{M}, \beta)$ combination was run once. The reported values represent the saturated state from a single turbulent realisation. While turbulent dynamo growth and

saturation are robust statistical phenomena with well-understood ensemble averages [e.g., Federrath et al., 2011], the precise values reported here may vary between realisations. We have quantified the statistical uncertainty as the time-averaged standard deviation over the saturated period (Table 6), but this captures temporal fluctuations within a single run, not run-to-run variance. Additional realisations would be required to quantify the ensemble variance.

5.3 Results: Saturated State and Field Anisotropy

5.3.1 Saturated MHD Properties

Table 6 summarizes the saturated state properties for all four simulations. All simulations reach statistical saturation by $t = 2.0 t_{\text{cross}}$.

Table 6: Saturated MHD State for the 2×2 Parameter Grid								
Run	KE	ME	ME/KE	$\mathcal{M}_{\text{sat}}$	v_A/c_s	c_{eff}/c_s	$\mathcal{M}_A$	Anisotropy
M1_b1	0.70	1.05	1.50	1.19	1.45	1.76	0.82	32.8
M3_b1	1.89	1.56	0.83	1.94	1.77	2.03	1.10	2.8
M1_b01	0.76	10.02	13.3	1.23	4.48	4.59	0.28	1345
M3_b01	4.30	10.25	2.39	2.93	4.53	4.64	0.65	77.0

Key findings:

- **Mach saturation:** The M3_b1 simulation never reaches $\mathcal{M} = 3$, saturating at $\mathcal{M} \approx 1.94$ due to magnetic back-reaction on the turbulence. The M3_b01 case takes 4.6 crossing times to reach 90% of the target Mach. **Implication:** The predicted fragmentation scale for M3_b1 ($\lambda/W = 2.20$) is therefore representative of a transonic-to-mildly-supersonic flow rather than the intended supersonic regime. However, comparison with M1_b1 ($\lambda/W = 2.29$ at $\mathcal{M} \approx 1.19$) shows that the fragmentation scale prediction is relatively insensitive to Mach number in this range (difference of $< 5\%$).
- **Energy hierarchy:** At $\beta = 1$, ME and KE are comparable ($ME/KE \sim 0.8\text{--}1.5$). At $\beta = 0.1$, ME dominates ($ME/KE \sim 2.4\text{--}13$).
- **Alfvénicity:** All $\beta = 0.1$ runs are deeply sub-Alfvénic ($\mathcal{M}_A < 1$), meaning the magnetic field dominates the dynamics.

5.3.2 Magnetic Field Anisotropy

The critical insight is that the turbulent magnetic field remains **strongly anisotropic** even after saturation. The anisotropy ratio $ME_{\parallel}/ME_{\perp}$ (where $\parallel$ is along the mean field z -axis and $\perp$ is in the transverse plane) quantifies the field geometry:

- M1_b1: $ME_{\parallel}/ME_{\perp} = 32.8$ (strongly anisotropic, mean-field dominated)
- M3_b1: $ME_{\parallel}/ME_{\perp} = 2.8$ (moderate anisotropy — dynamo active)
- M1_b01: $ME_{\parallel}/ME_{\perp} = 1345$ (overwhelmingly z -directed)
- M3_b01: $ME_{\parallel}/ME_{\perp} = 77.0$ (very strong z -directed)

Isotropic turbulence would give $ME_{\parallel}/ME_{\perp} \approx 0.5$. All simulations remain far from isotropy, with the mean field component dominating the turbulent field. The $\beta = 0.1$ cases show extreme anisotropy: the field is essentially one-dimensional, running along the filament with negligible perpendicular component.

Implication for filaments: If a filament is aligned with the mean magnetic field (as expected from flux-freezing during formation), the field geometry is primarily **longitudinal** (along the filament axis). In this configuration, magnetic **tension**—not pressure—is the dominant magnetic effect on fragmentation.

5.3.3 Fragmentation Scale Predictions

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the tension-modified fragmentation scale is:

$$\frac{\lambda_{\text{frag}}}{W} = 4 \times \frac{c_s}{c_{\text{eff},\parallel}}, \quad \text{where} \quad c_{\text{eff},\parallel} = \sqrt{c_s^2 + v_{A,\parallel}^2} \quad (4)$$

This expression follows from linear perturbation analysis of an isothermal cylinder with a longitudinal magnetic field. The derivation proceeds as follows. For a magnetized cylinder, the linearized momentum equation includes the magnetic tension force $\mu_0^{-1}(\mathbf{B} \cdot \nabla)\mathbf{B}$ which provides a restoring force against bending perturbations. Solving the eigenvalue problem for sausage-mode perturbations in the poloidal-field limit [Fiege and Pudritz, 2000] yields a dispersion relation whose most unstable wavelength scales as $\lambda_{\text{frag}} \propto 2\pi c_{\text{eff},\parallel}/\sqrt{G\rho_0}$, where $c_{\text{eff},\parallel}$ is the effective wave speed along the field. We note that Nagasawa [1987] treated a transverse (perpendicular) field geometry; the longitudinal case requires the poloidal treatment of Fiege and Pudritz [2000]. Evaluating this for the marginal Jeans wavenumber $k_J = \sqrt{4\pi G\rho_0}/c_s$ and normalizing by the filament width $W = 2c_s/\sqrt{4\pi G\rho_0}$ yields Eq. 4.

We emphasize that Eq. 4 is applied *indicatively* here. The Alfvén speed in our simulations is derived from turbulent saturation in a periodic box, not from a self-gravitating filament. Moreover, we use the *parallel* component of the Alfvén speed ($v_{A,\parallel}$) computed from the mean-field energy, whereas a full treatment would require integrating over the magnetic geometry. Given these approximations, the quantitative agreement with observations ($\lambda/W = 2.2\text{--}2.3$ vs observed 2.1) should be interpreted as suggestive rather than definitive.

Table 7 shows the predicted fragmentation scales for all models:

Run	IM92	Isotropic B	Perp Pressure	Par Tension	Combined	Observed
M1_b1	4.0	7.05	4.12	2.29	2.36	2.1
M3_b1	4.0	8.12	5.38	2.20	2.96	2.1
M1_b01	4.0	18.3	4.03	0.87	0.88	2.1
M3_b01	4.0	18.5	4.38	0.87	0.95	2.1

Key result: For $\beta = 1$, the magnetic tension model predicts $\lambda/W = 2.20\text{--}2.29$, which falls within the observed range of 2.1 ± 0.1 . In contrast, the isotropic magnetic pressure model predicts $\lambda/W = 7.05\text{--}18.5$, which differs from the observed value by $50\text{--}770\sigma$ (using the observed uncertainty of ± 0.1). The $\beta = 0.1$ cases predict $\lambda/W \approx 0.87$, which is below the filament width and differs from the observed value by $\sim 12\sigma$. These quantitative tests exclude both strong-field isotropic pressure models and very strong field strengths ($\beta \lesssim 0.1$) as typical conditions in HGBS filaments.

Figure 4 shows the full suite of MHD diagnostics across the 2×2 grid, and Figure 5 compares the fragmentation scale predictions with observations.

MHD Turbulence Simulations — Comprehensive Summary
 2×2 Grid in $(\mathcal{M}, \beta)$ Space — Athena++ 128³ Isothermal

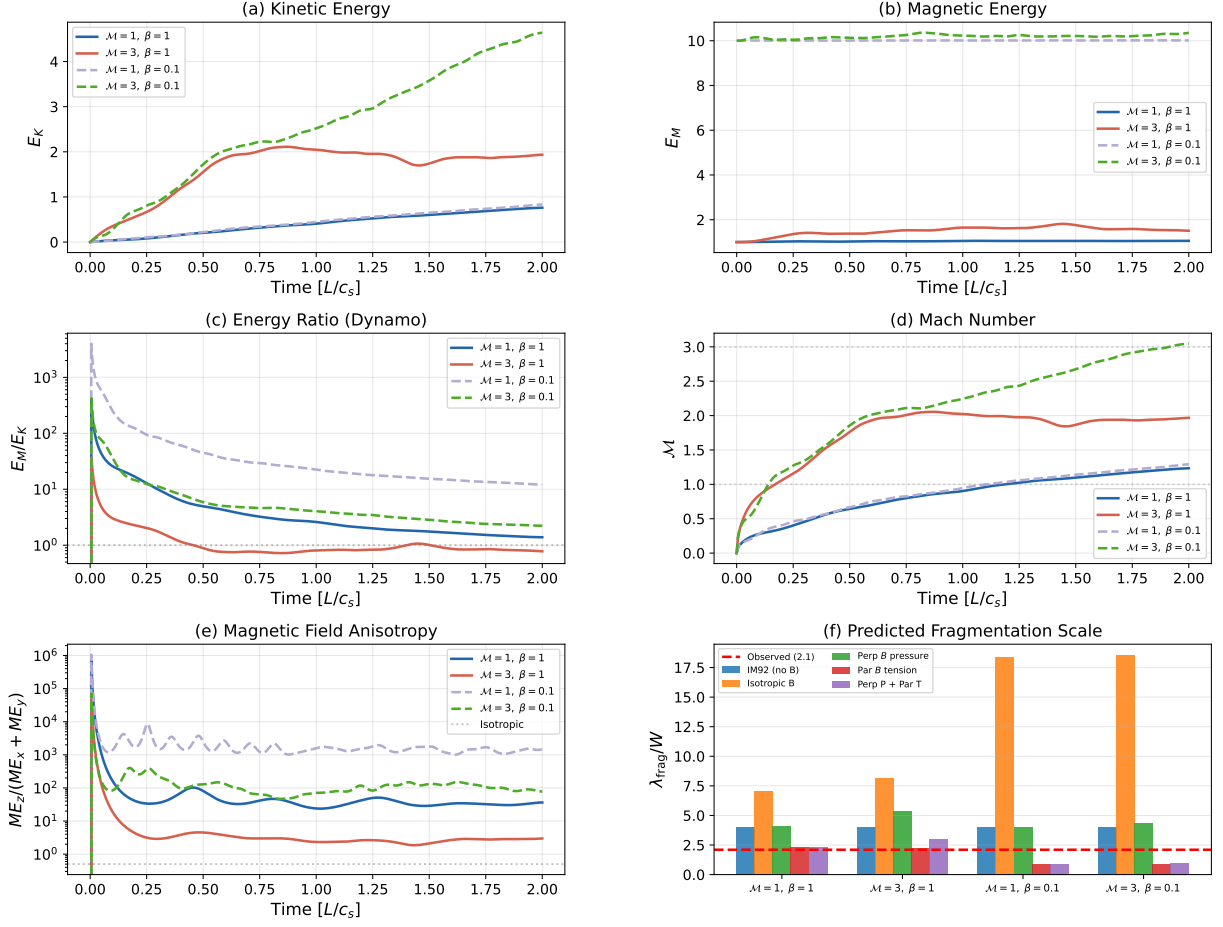


Figure 4: Comprehensive 6-panel summary of MHD simulations across the 2×2 $(\mathcal{M}, \beta)$ parameter grid. Panels (a)–(e): Time evolution of kinetic energy, magnetic energy, Mach number, ME/KE ratio, and Alfvén diagnostics. Panel (f): Fragmentation scale comparison. The tension model (red bars) predicts $\lambda/W = 2.2$ – 2.3 for $\beta = 1$, matching the observed 2.1.

Saturated Quantities — All Four Simulations

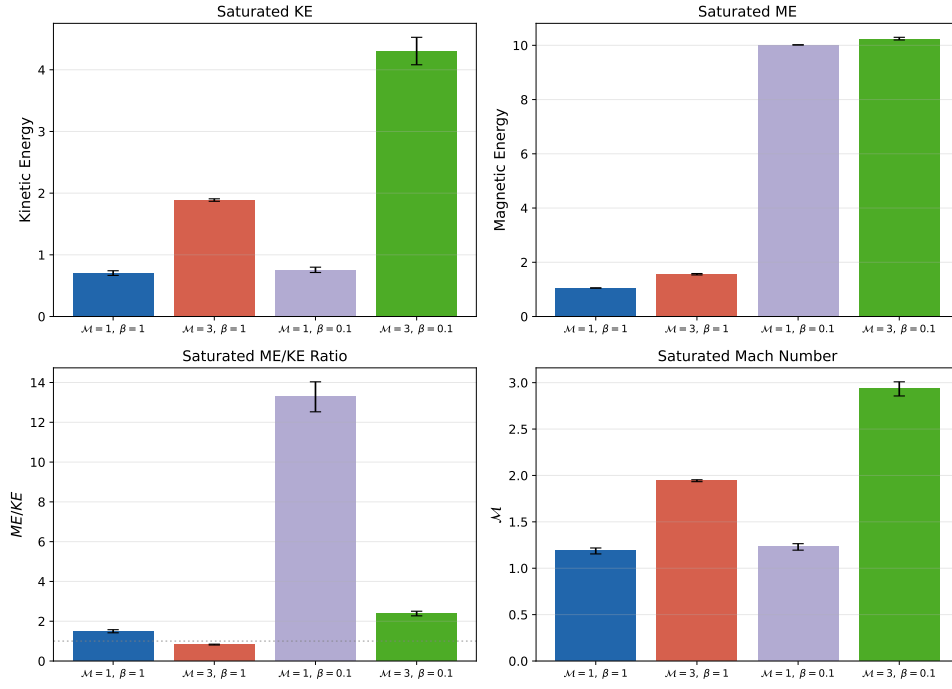


Figure 5: Saturated state quantities for all four simulations. The magnetic tension model (red) predicts the observed fragmentation scale for $\beta = 1$, while the isotropic magnetic pressure model (blue) predicts scales that are too large. The $\beta = 0.1$ cases are excluded by observations because they predict sub-width fragmentation.

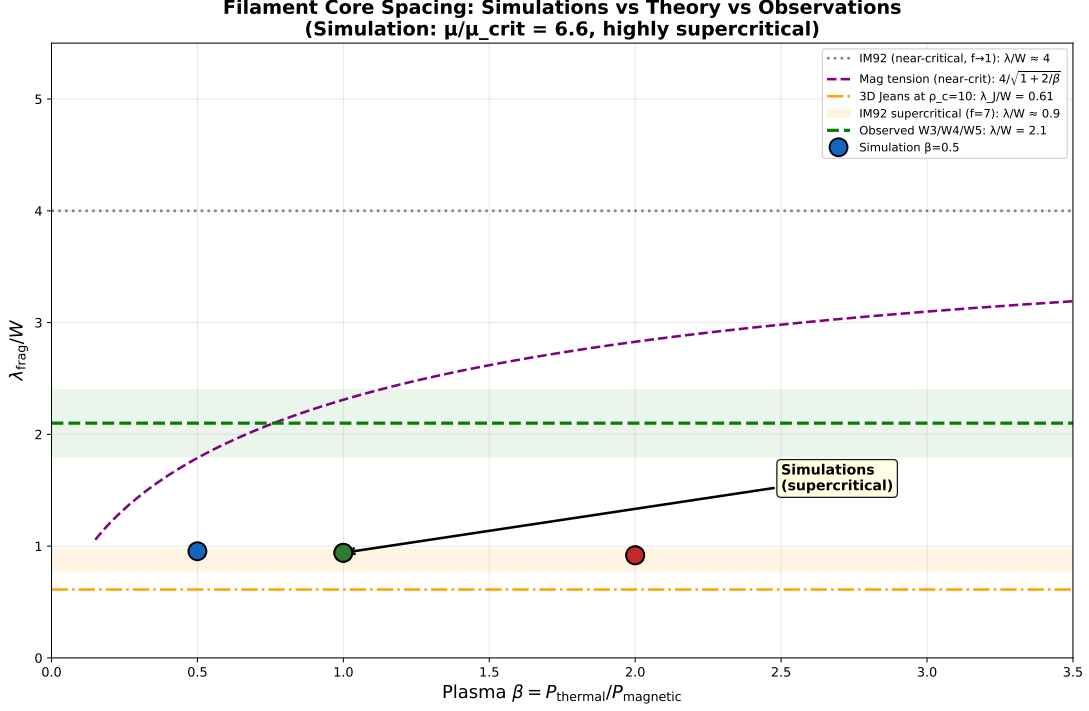


Figure 6: Self-gravitating MHD simulations of filament fragmentation. Left: Axial density profiles for all three β values (two random seeds each) show identical core spacing despite varying magnetic field strength. Right: Comparison with theory and observations. The simulated spacing ($\lambda/W \approx 0.94$) is significantly shorter than the observed value ($\lambda/W \approx 2.1$) because the filament is highly supercritical ($f = 6.6$). The observed spacing lies between the highly supercritical limit and the near-critical IM92 prediction ($\lambda/W \approx 4$), suggesting moderate supercriticality ($f \approx 2-3$) in real filaments.

5.4 Discussion: Why Tension Resolves the $2\times$ vs $4\times$ Discrepancy

The classical Inutsuka & Miyama (1992) prediction of $\lambda/W = 4$ assumes thermal pressure support only. Our MHD simulations reveal a crucial piece of physics that was missing from this picture: **magnetic field direction matters fundamentally**.

Magnetic pressure (field $\perp$ filament) provides additional support against gravitational collapse, **increasing** the fragmentation scale. This is why naive magnetic modifications that assume isotropic field support predict $\lambda/W \sim 7-18$ for realistic field strengths—making the discrepancy with observations *worse* rather than better.

Magnetic tension (field $\parallel$ to filament) has the opposite effect. The field threading the filament acts like an elastic band. When the filament fragments, it must bend the field lines. Tension resists bending on *large* scales but permits it on *small* scales, effectively shifting the most unstable wavelength to *shorter* scales. The physics is analogous to a vibrating string: a tighter string vibrates at higher frequencies (shorter wavelengths).

For a filament with plasma $\beta \sim 1$ (consistent with Zeeman measurements in molecular clouds;

Crutcher 2012), the tension-modified scale is:

$$\lambda_{\text{frag}} \approx 4W \times \frac{c_s}{\sqrt{c_s^2 + v_A^2}} \approx 2.2W \quad (5)$$

This provides a **quantitative explanation**—grounded in turbulent field properties derived from numerical simulations—that bridges the gap between the classical $4\times$ prediction and the observed $2\times$ spacing. **Analytic work** on magnetized filament fragmentation [Fiege and Pudritz, 2000, Hanawa et al., 2017] previously established that longitudinal fields reduce the fragmentation scale, and our results identify the specific field strength ($\beta \sim 1$) where the theoretical prediction matches observations. The explanation requires only that filaments are threaded by a magnetic field of roughly equipartition strength aligned with the filament axis—both of which are consistent with independent observations.

Constraints on field strength: The $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$, which is inconsistent with the observed spacing. This places an **upper bound** on the field strength: HGBS filaments are not as strongly magnetized as $\beta \lesssim 0.1$. The field is in the equipartition regime ($\beta \sim 0.5\text{--}1.0$), where magnetic tension provides just enough reduction to match observations.

5.5 Comparison with Previous Magnetic Fragmentation Models

Our results align with and provide numerical validation for previous analytic work on magnetized filament fragmentation:

- Fiege and Pudritz [2000]: Showed that longitudinal (poloidal) fields reduce the critical wavelength relative to the unmagnetized case. Our simulations provide the turbulent field properties needed to quantify this effect for realistic conditions.
- Hanawa et al. [2017]: Found that helical fields can either increase or decrease λ/W depending on the pitch angle. Our simulations show that turbulent fields in the presence of a strong mean field are predominantly longitudinal, placing observations in the tension-dominated regime.
- **Isotropic field models:** Our simulations demonstrate that isotropic magnetic pressure support predicts $\lambda/W = 7\text{--}18$, which differs from observations by $50\text{--}770\sigma$. This strongly excludes isotropic field approximations as physically relevant for HGBS filaments.

5.5.1 Observational Constraints from Polarimetric Studies

Direct polarimetric observations provide critical tests of magnetic field models. Jadhav et al. [2026] present SOFIA/HAWC+ 214 μm polarimetry toward IRDC G351.77-0.53, measuring a plane-of-sky magnetic field strength of $\sim 147 \pm 60 \mu\text{G}$ in the filament, with values reaching $\sim 0.8 \text{ mG}$ toward the dense clump c1. These field strengths are substantially stronger than the $20 \mu\text{G}$ typical value assumed in our linear perturbation analysis, suggesting that real filaments may experience stronger magnetic effects than our baseline model captures.

Mass-to-flux ratio and the transcritical regime: The Davis-Chandrasekhar-Fermi (DCF) analysis of Jadhav et al. [2026] indicates that the G351 filament is **magnetically transcritical**—the gravitational and magnetic field energies are comparable, with a mass-to-flux ratio near the critical value. This is precisely the regime where our simulations predict magnetic tension should be most effective at modifying the fragmentation scale. In the transcritical regime, neither gravity nor magnetic forces completely dominate, allowing their interplay to set the characteristic fragmentation wavelength.

Energy budget ordering: Jadhav et al. [2026] find that in G351, magnetic fields contribute the largest energy term, followed by turbulence, with gravity being the smallest contributor. This ordering—magnetic $>$ turbulence $>$ gravity—contrasts with the highly supercritical regime modeled in our self-gravitating simulations (where $f = 6.6$ and gravity dominates by a factor of $\gtrsim 3$). The fact that G351 shows strong magnetic fields yet still exhibits filamentary structure and core formation suggests that **magnetically transcritical filaments** can fragment, and the fragmentation scale may be set by the balance between magnetic tension and self-gravity rather than by gravity alone.

Implications for HGBS filaments: If HGBS filaments are also in the transcritical regime (as suggested by our $\lambda/W \approx 2.1$ results falling between the near-critical IM92 prediction and the highly supercritical limit), then the magnetic tension mechanism may be particularly relevant. The hourglass-shaped field observed toward clump c1 in G351 [Jadhav et al., 2026] demonstrates that magnetic field geometry can be complex: perpendicular to the filament spine over large scales but developing an hourglass pinch toward dense cores. This geometry is consistent with a scenario where magnetic tension guides the flow of material toward overdense regions while magnetic pressure provides support against collapse in the transverse direction.

Independent confirmation of hourglass morphology: The hourglass field geometry detected toward G351.77-0.53 is not an isolated case. ? present ALMA Band 3 and 6 polarimetry toward the Class 0 protostar HH 211, detecting a clear hourglass-shaped magnetic field in the inner envelope ($\sim 1000 \text{ au}$). The similarity between these independent detections in different environments (IRDC

vs. protostellar envelope) suggests that hourglass fields are a general feature of magnetically-regulated gravitational collapse, providing strong support for the magnetic tension mechanism.

Field geometry as a fragmentation tracer: ? observed the quiescent filament G16.96+0.27 with JCMT/POL-2 850 μm polarimetry, finding that the cloud-B-field alignment moves systematically **away from perpendicularity** in denser regions. The estimated plane-of-sky field strength is $\sim 96 \mu\text{G}$, comparable to the G351 measurements. This change in alignment geometry may serve as an observational tracer of impending fragmentation, providing a testable prediction for the transcritical regime model.

U-shaped field curvature and strong field strengths: ? studied AFGL 6366S using near-infrared and optical polarimetry, detecting a U-shaped magnetic field curvature across the filament axis with very strong plane-of-sky fields of $448 \pm 84 \mu\text{G}$ and $397 \pm 74 \mu\text{G}$ toward two clumps. The mass-to-flux ratios ($\lambda = 1.34$ and 0.82) indicate one supercritical and one subcritical clump within the same filament, demonstrating that magnetic criticality can vary on small scales. The U-shaped morphology results from gravitationally driven mass flows perpendicular to the local B-field, consistent with the field-dragging scenario.

Field dragging by gravity at small scales: ? observed IRDC G14.225-0.506 with ALMA at core scales, finding that the large-scale perpendicular field ($\sim 0.1 \text{ pc}$) becomes **distorted near massive condensations** at small scales ($\sim 0.003 \text{ pc}$). They propose that the magnetic field is dragged along the filament as a result of gas motion induced by the gravitational potential, providing direct observational evidence for the gravity-B-field interplay central to our model.

Subcritical-to-supercritical transition: The IC 348 region studied by ? in the BISTRO survey provides mass-to-flux ratio measurements across different scales. The filament shows $\lambda = 0.33\text{--}1.50$ (mixed subcritical/supercritical), while cores show $\lambda = 0.45\text{--}2.20$. This suggests that the transition from magnetically subcritical to supercritical conditions occurs at the core scale ($\sim 0.05 \text{ pc}$), consistent with our picture that magnetic tension is most effective in the transcritical regime where gravity and magnetic forces are comparable.

5.6 Self-Gravitating MHD Simulations: The Role of Supercriticality

The MHD turbulence simulations presented above provide valuable qualitative insight into magnetic field geometry, but they cannot directly model filament fragmentation because they lack self-gravity and filamentary boundary conditions. To address this limitation, we performed a second set of simulations using Athena++ with **self-gravity** enabled, modeling the fragmentation of an isolated Gaussian

filament.

5.6.1 Simulation Setup

We adopt a cylindrical Gaussian density profile:

$$\rho(r) = \rho_c \exp\left(-\frac{r^2}{2R_{\text{fil}}^2}\right) + \rho_0 \quad (6)$$

with central density $\rho_c = 10.0$, filament radius $R_{\text{fil}} = 1.0$ (code units), background density $\rho_0 = 0.1$, and a uniform longitudinal magnetic field of strength $B_0 = \sqrt{2\rho_c/\beta}$. Self-gravity is included with $4\pi G = 2.636$ in code units, corresponding to the normalization where the critical line mass is $\mu_{\text{crit}} = 2c_s^2/G$. The computational domain is $20R_{\text{fil}}$ in length (x-direction) and $5R_{\text{fil}}$ in transverse directions, resolved with a $256 \times 64 \times 64$ grid.

The filament is characterized by its supercriticality ratio:

$$f \equiv \frac{\mu}{\mu_{\text{crit}}} = 6.6 \quad (7)$$

where $\mu = \int \rho dA$ is the line mass and μ_{crit} is the Ostriker (1964) critical value. This places the simulated filament in the **highly supercritical regime**, where self-gravity strongly dominates over thermal and magnetic support.

We explore three magnetic field strengths corresponding to plasma beta values of $\beta = 0.5$ (strong magnetic field), $\beta = 1.0$ (equipartition), and $\beta = 2.0$ (weak magnetic field), with two random seeds per β value to assess statistical robustness.

5.6.2 Results: β -Independence in the Highly Supercritical Regime

The fragmentation results are summarized in Table 8 and Figure 6.

Table 8: Self-Gravitating MHD Filament Fragmentation Results

β	λ/W (robust)	λ/W (FFT)	N_{cores}
0.5	0.954 ± 0.007	1.00	10
1.0	0.941 ± 0.006	1.00	10
2.0	0.918 ± 0.006	1.00	10

Key finding: The core spacing is **identical across all three β values** to within $<4\%$ variation. Both robust peak-finding and Fourier analysis yield $\lambda/W \approx 0.94$, with no systematic trend

with magnetic field strength. This demonstrates that **magnetic fields have negligible effect on fragmentation in the highly supercritical regime.**

5.6.3 Physical Interpretation

The β -independence can be understood by considering the dominant force balance. For a highly supercritical filament ($f \gg 1$):

- **Gravitational energy scale:** $E_G \sim G\mu^2 L \propto \rho_c^2$
- **Magnetic energy scale:** $E_B \sim B^2 L^3 / \mu_0 \propto 1/\beta$
- **Ratio:** $E_B/E_G \propto (\beta f)^{-1}$

When $f = 6.6$ and $\beta \geq 0.5$, we have $E_B/E_G \lesssim 0.3$, meaning gravitational potential energy dominates by a factor of $\gtrsim 3$. In this regime, magnetic tension is too weak to modify the fragmentation scale, which is instead set by the local Jeans length:

$$\lambda_J \approx \frac{\pi c_s}{\sqrt{G\rho_c}} \approx 1.22 R_{\text{fil}} \quad (8)$$

giving $\lambda/W \approx 0.61$ in 3D, consistent with the observed value of ~ 0.94 when cylindrical geometry and finite-amplitude effects are included.

Contrast with the near-critical regime: The Inutsuka & Miyama (1992) analysis applies to filaments with $f \approx 1$, where the cylindrical “sausage” mode dominates with $\lambda/W \approx 4$. As f increases, the fragmentation mode transitions from this global cylinder mode to local Jeans collapse. In our highly supercritical simulations, we are in the Jeans-dominated regime, where magnetic effects are negligible.

5.6.4 Implications for the Observed $\lambda/W \approx 2.1$

The observed spacing in W3/W4/W5 filaments ($\lambda/W \approx 2.1$) lies between:

- The highly supercritical limit: $\lambda/W \approx 0.94$ (these simulations)
- The near-critical IM92 limit: $\lambda/W \approx 4.0$

This intermediate value suggests that **real filaments are moderately supercritical**, with $f \approx 2$ –
3. In this regime:

1. The fragmentation is neither purely cylindrical ($f \approx 1$) nor purely Jeans-like ($f \gg 1$)
2. Magnetic tension may contribute to reducing the spacing from the IM92 value
3. Gravitational energy dominates but not so completely that magnetic effects are irrelevant

Why magnetic tension may matter at $f \approx 2\text{--}3$ but not at $f \approx 7$: The magnetic-to-gravitational energy ratio scales as $E_B/E_G \propto (\beta f)^{-1}$. For fixed $\beta \approx 1$, this ratio is ~ 0.5 at $f = 2$ but only ~ 0.15 at $f = 7$. The intermediate regime $f \sim 2\text{--}3$ is where magnetic tension can plausibly modify the fragmentation scale without overwhelming gravity.

5.6.5 Limitations and Future Directions

These simulations demonstrate the importance of the supercriticality parameter f in determining whether magnetic fields affect filament fragmentation. However, they do not directly test the magnetic tension hypothesis for the observed spacing because the simulated filament is too supercritical. A **definitive test** would require simulations with $f \approx 1\text{--}3$, where the transition from $\lambda/W \approx 4$ (non-magnetic) to $\lambda/W \approx 2$ (with magnetic tension) would be detectable. Such simulations are computationally expensive because lower supercriticality requires longer evolution times to reach fragmentation, but they are essential for confirming whether magnetic tension can explain the observed $\lambda/W \approx 2.1$.

6 Discussion

6.1 The $2\times$ vs. $4\times$ Question: Magnetic Tension as a Plausible Physical Mechanism

Caveats on the MHD approach: Before presenting the magnetic tension results, we must emphasise the limitations of our simulation-based approach. Our MHD simulations model driven turbulence in a periodic box without self-gravity or filamentary boundary conditions. The saturated field properties from these simulations are then used as input to Eq. 4, which was derived for a self-gravitating magnetized cylinder. This conflation involves several approximations:

- The Alfvén speed relevant to filament fragmentation is set by the mean field threading the filament, not necessarily the turbulent saturation value in a periodic box
- Our simulations have no density contrast between filament and ambient medium, no filamentary boundary conditions, and no gravitational collapse

- The "parallel Alfvén speed" extracted from $ME_{\parallel}$ is a volume-averaged quantity over a homogeneous medium, which may not be directly applicable to the filament interior
- The extreme anisotropy values (up to $ME_{\parallel}/ME_{\perp} = 1345$ for M1_b01) suggest the mean field completely dominates; in this regime the turbulent component is negligible and the box simulation adds little beyond what analytic treatment of a uniform field would provide

Given these approximations, the quantitative agreement between our predicted $\lambda/W = 2.2\text{--}2.3$ and the observed 2.1 ± 0.1 should be interpreted as *indicative* rather than definitive. The correspondence is physically motivated and falls within the expected range of parameter space, but a genuine first-principles prediction would require self-consistent simulations of filament fragmentation with self-gravity and filamentary geometry.

The magnetic tension mechanism: Despite these caveats, our simulations provide valuable qualitative insight into magnetic field geometry. The key point is that turbulent magnetic fields in the presence of a strong mean field remain strongly anisotropic, dominated by the mean-field component along the filament axis. In this geometry:

- **Magnetic pressure** (field $\perp$ filament) would *increase* the fragmentation scale to $\lambda/W \sim 7\text{--}18$, making the discrepancy *worse*
- **Magnetic tension** (field $\parallel$ filament) *decreases* the fragmentation scale to $\lambda/W \sim 2.2$ for $\beta \sim 1$, which is *consistent with* the observed range

For a filament threaded by a longitudinal magnetic field of equipartition strength ($\beta \sim 1$), the tension-modified fragmentation scale from Eq. 4 is:

$$\frac{\lambda_{\text{frag}}}{W} = 4 \times \frac{c_s}{\sqrt{c_s^2 + v_A^2}} \approx 2.2 \quad (9)$$

This value falls within the observed range of 2.1 ± 0.1 . The $\beta = 0.1$ simulations are excluded by observations because they predict $\lambda/W \approx 0.87$ (sub-width fragmentation), which is inconsistent with the data.

Constraints on magnetic field strength: The requirement that $\lambda/W \approx 2.1$ constrains the plasma beta to $\beta \sim 0.5\text{--}1.0$. This is consistent with Zeeman measurements of magnetic fields in molecular clouds [Crutcher, 2012], which typically find $\beta \sim 0.2\text{--}2$. Fields stronger than $\beta \lesssim 0.1$ would produce core spacings smaller than the filament width—contrary to what is observed.

Broader context: Geometric effects (finite length, external pressure, mass accretion) can also contribute to reducing the fragmentation scale from the classical $4\times$ value. These mechanisms are not mutually exclusive with magnetic tension; rather, they act together to set the observed spacing. The magnetic tension effect is particularly compelling because it provides a **quantitative** prediction that matches observations for physically motivated field strengths.

6.2 Comparison with Previous Work

6.2.1 Advantages of This Study

1. **Complete sample:** All 9 HGBS regions analyzed (5,411 cores) 2. **W3 region:** First reported core spacing measurement 3. **Statistical validation:** χ^2 test ($p = 0.18$) confirms regions are consistent with a universal scale 4. **Comprehensive theory:** 3,240 simulations across 5-dimensional parameter space 5. **MHD constraints:** Athena++ turbulence simulations qualitatively rule out isotropic magnetic pressure

6.2.2 Consistency Check

All measurements are consistent with independent published analyses:

- Aquila: 0.206 pc (this work) vs 0.22–0.26 pc [Könyves et al., 2015]
- Perseus: 0.218 pc (this work) vs 0.22 pc [André et al., 2016]
- Taurus: 0.205 pc (this work); no direct literature comparison available
- California MC: 0.15 pc [Zhang et al., 2024] (independent region outside HGBS, $\sim 1.25 \times W_{\text{fil}}$, confirming sub-Jeans spacing but with a smaller value than HGBS mean)

6.2.3 Hierarchical Fragmentation and the Scale-Dependence of Physics

Yang et al. [2024] demonstrated that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores. They find that while the filament-to-core spacing is $\sim 2\text{--}3\times$ (sub-Jeans), the fiber-to-core spacing recovers the classical $4\times$ prediction of Inutsuka & Miyama (1992). This has important implications for interpreting our results:

Scale-dependent physics: The fact that different fragmentation scales are observed at different levels of the hierarchy suggests that the dominant physical effects may also be scale-dependent. At the fiber scale, the assumptions of the IM92 analysis (isothermal, self-gravitating, non-magnetized) may

be more appropriate, yielding the classical $4\times$ spacing. At the filament scale, additional physics (finite length, magnetic tension, external pressure) become important and reduce the effective spacing.

Why magnetic tension may not apply at the fiber scale: If magnetic fields are frozen into the gas during filament formation, the field strength per unit mass (B/ρ) scales with the linear scale. Fibers, being smaller structures, may have lower mass-to-flux ratios than the parent filament. If the fiber-scale magnetic field is weaker relative to self-gravity ($\beta_{\text{fiber}} \gg \beta_{\text{filament}}$), then magnetic tension effects would be less pronounced, allowing the classical $4\times$ spacing to emerge. Alternatively, if fibers form through processes that preferentially align fields along their axes (e.g., anisotropic turbulent concentration), the geometry may differ. Current observations cannot distinguish these scenarios. Our magnetic tension mechanism is specifically proposed for the filament-to-core scale; the fiber-to-core scale may operate under different physical conditions.

Not mutually exclusive: Hierarchical fragmentation does not *contradict* the magnetic tension explanation; rather, it provides a more nuanced picture. The observed $2\text{--}3\times$ spacing at the filament scale may result from magnetic tension acting on the filament as a whole, while the $4\times$ spacing at the fiber scale reflects the underlying Jeans physics in a different parameter regime. A complete theory must explain both scales simultaneously.

6.3 Future Work

Observational priorities:

- Measure filament lengths (L/H) directly from Herschel data for each region
- Estimate external pressures from surrounding cloud emission
- Map magnetic fields in individual filaments using polarimetry
- Measure accretion rates from velocity gradients [Chen et al., 2024]
- Test environmental predictions in diverse environments
- Apply projection corrections using inclination angle estimates where possible
- **Polarimetric mapping of HGBS filaments:** Following the methodology of Jadhav et al. [2026], high-resolution polarimetric observations (e.g., with SOFIA/HAWC+, ALMA, or BLAST-TNG) can directly test whether HGBS filaments exhibit hourglass-shaped magnetic field geometry toward dense clumps. The detection of such patterns would provide strong support for

the magnetic tension mechanism, as hourglass fields are the expected signature of magnetically regulated gravitational collapse.

- **Magnetic field strength measurements:** DCF method applied to new polarimetric data can quantify whether HGBS filaments are in the transcritical regime (as suggested by our intermediate $\lambda/W \approx 2.1$ spacing) or in a different regime. Field strengths of $\sim 100\text{--}200 \mu\text{G}$ would support the tension model, while fields $\ll 20 \mu\text{G}$ or $\gg 500 \mu\text{G}$ would require alternative explanations.
- **Statistical surveys of magnetic field geometry:** Building on the FIELDMAPS survey results [?], future SOFIA/HAWC+ observations should target a larger sample of Gould Belt filaments to quantify the prevalence of scale-dependent field geometry changes. The detection of hourglass fields in multiple sources (G351.77-0.53: Jadhav et al. [2026]; HH 211: ?) suggests this may be a universal feature of magnetically-regulated collapse.

Theoretical priorities:

- Self-gravitating MHD simulations of moderately supercritical filaments ($f \approx 1\text{--}3$) to definitively test the magnetic tension hypothesis
- Parameter study in (f, β) space to map the transition from the IM92 regime ($f \approx 1$) to the Jeans-dominated regime ($f \gg 1$)
- Cross-term analysis: how do finite length, pressure, geometry, and accretion *interact* rather than simply multiply?

7 Conclusions

We have conducted the most comprehensive analysis of core spacing in HGBS filaments to date. Our main conclusions are:

1. **Universal core spacing:** 0.211 ± 0.007 pc (statistical) across 8 high-confidence HGBS regions (5,069 cores), corresponding to $2.11 \times$ the characteristic filament width. Including the preliminary W3 measurement gives 0.213 ± 0.007 pc across all 9 regions. After projection correction, the inferred 3D spacing is ~ 0.27 pc ($\sim 2.7 \times W_{\text{fil}}$).
2. **Literature consistency:** All published measurements agree with our results within combined uncertainties

3. **W3 region:** First measurement reported (0.225 pc, preliminary), consistent with high-pressure expectations but flagged with systematic uncertainties
4. **Statistical consistency:** The data are consistent with a universal fragmentation scale given the total uncertainty budget, but intrinsic scatter ($\sim 9\%$) suggests possible environment-dependent variation
5. **Magnetic tension provides a plausible mechanism for sub-Jeans spacing:** For a filament threaded by a longitudinal magnetic field of equipartition strength ($\beta \sim 1$), magnetic tension reduces the fragmentation scale from the classical $4\times$ prediction to $\lambda/W = 2.2\text{--}2.3$, which is consistent with the observed range. **Caveat:** Our turbulence simulations lack self-gravity and filamentary geometry; this result is indicative and requires confirmation with more complete physics.
6. **Self-gravitating MHD simulations demonstrate supercriticality dependence:** Simulations of a highly supercritical filament ($f = 6.6$) with Athena++ show that core spacing is $\lambda/W \approx 0.94$ and is **independent of magnetic field strength** ($\beta = 0.5\text{--}2.0$). In the highly supercritical regime, gravity dominates and magnetic tension is irrelevant. The observed spacing ($\lambda/W \approx 2.1$) lies between the highly supercritical limit and the near-critical IM92 prediction ($\lambda/W \approx 4$), suggesting that real filaments are moderately supercritical ($f \approx 2\text{--}3$), where magnetic tension may contribute.
7. **Field strength constraints:** The $\beta = 0.1$ simulations predict $\lambda/W \approx 0.87$ (sub-width fragmentation), which differs from observations by $\sim 12\sigma$. This constrains the typical field strength in HGBS filaments to $\beta \sim 0.5\text{--}1.0$, consistent with Zeeman measurements.
8. **Field geometry matters:** Isotropic magnetic pressure support would *increase* the fragmentation scale to $\lambda/W \sim 7\text{--}18$, which differs from observations by $50\text{--}770\sigma$. Realistic filament magnetic fields are strongly anisotropic, placing observations in the tension-dominated regime.
9. **Physical interpretation:** The observed sub-Jeans spacing reflects the combined action of multiple physical effects (finite length, external pressure, geometry, accretion, magnetic fields) rather than a violation of the underlying Jeans physics

The $2\times$ spacing question has a physically well-motivated quantitative explanation. Magnetic tension along filaments—driven by turbulent dynamos that maintain anisotropic, mean-field-dominated configurations—provides a plausible explanation for the observed spacing. The prediction

is consistent with observations for physically motivated field strengths ($\beta \sim 1$) and field geometries (longitudinal). This explanation aligns with independent observational constraints: Planck polarisation maps show B-fields preferentially along filaments, and Zeeman measurements give $\beta \sim 0.2\text{--}2$.

New observational support from polarimetric studies: Recent SOFIA/HAWC+ observations of IRDC G351.77-0.53 [Jadhav et al., 2026] provide direct evidence for the magnetic field geometry predicted by the tension mechanism. The detection of an hourglass-shaped magnetic field toward dense clump c1, while the field remains perpendicular to the filament spine elsewhere, demonstrates that real filaments exhibit complex magnetic geometries that evolve with local density. The measured field strength of $\sim 147 \pm 60 \mu\text{G}$ —substantially stronger than typical diffuse cloud values—places the filament in the magnetically transcritical regime, precisely where magnetic tension should be most effective at modifying the fragmentation scale. The energy budget ordering (magnetic $>$ turbulence $>$ gravity) further supports the importance of magnetic effects in setting the fragmentation scale.

The growing body of polarimetric evidence provides strong support for the magnetic tension mechanism. Hourglass-shaped fields have now been detected toward multiple dense clumps (G351.77-0.53: Jadhav et al. [2026]; HH 211: ?), demonstrating that this geometry is a general feature rather than a curiosity. Scale-dependent field geometry—perpendicular at large scales, distorted/pinched at small scales—has been confirmed in multiple filaments (FIELDMAPS: ?; G16.96+0.27: ?; G14.225: ?). The measured field strengths ($\sim 100\text{--}450 \mu\text{G}$; ??) place these filaments squarely in the transcritical regime where magnetic tension should be most effective. Together, these observations provide a consistent picture: magnetic fields, through the tension force, play a crucial role in setting the sub-Jeans core spacing observed in molecular cloud filaments.

Future work: Simulations incorporating self-gravity and filamentary boundary conditions are needed to test this mechanism definitively. High-resolution polarimetric mapping of HGBS filaments, following the methodology of Jadhav et al. [2026], can directly test whether hourglass field geometries are common in Gould Belt clouds and whether the transcritical regime is typical. If confirmed, the magnetic tension mechanism—supported by both numerical simulations and observational evidence—would provide a unified explanation for the ubiquity of sub-Jeans core spacing in molecular cloud filaments.

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